Quiz 9

Friday, November 6 products, powers, and computational cost (Week 10)

Last updated: October 19, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers the week before only: Products, Powers, and Cost (Oct 26 30). Same-day determinants and Lab 4 are **not** on this quiz.

Textbooks: Boyd & Vandenberghe (VMLS) also on Resources.

Schedule

Core

Fluency (complete prompts practice these by hand)

C9.1. Let $a = \begin{bmatrix}1\\2\\0\end{bmatrix}$, $b = \begin{bmatrix}1\\0\\-1\end{bmatrix}$, and $c = \begin{bmatrix}2\\1\\1\end{bmatrix}$.

(a) Form the outer product ab^T (give the 3×3 matrix).

(b) Compute $(ab^\mathsf{T})c$ and $a(b^\mathsf{T}c)$. Confirm they are equal as vectors.

C9.2. In C9.1, which evaluation order stores a full 3×3 intermediate matrix? Which only needs a scalar intermediate $b^\mathsf{T}c$? In one sentence, why does the second order win for large n when $a, b, c \in \mathbb{R}^n$?

C9.3. Let

$$A = \begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}, \qquad B = \begin{bmatrix}2 & 0\\0 & 1\end{bmatrix}, \qquad x = \begin{bmatrix}1\\0\end{bmatrix}.$$

Interpret ABx as apply B, then A. Compute Bx, then $A(Bx)$. Also compute BAx. Is $AB = BA$?

C9.4. Dimensions. For each pair, decide whether AB is defined and give the size of AB if it is.

(a) A is 4×2, B is 2×3.

(b) A is 3×2, B is 3×2.

(c) A is 2×2, B is 2×5.

C9.5. Compute

$$\begin{bmatrix}1 & 2\\0 & -1\end{bmatrix}\begin{bmatrix}0 & 1\\3 & 2\end{bmatrix}$$

by forming columns Ab_1 and Ab_2 (or by row column inner products).

C9.6. Stage 1: $B = \mathrm{diag}(2, 1)$. Stage 2: $A = \begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}$ (90° CCW). Write the single matrix M for B then A and compute $M(1, 1)$.

C9.7. Let $A = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ (swap coordinates) and $x_0 = (4, 1)$. Compute A^2 and $x_2 = A^2x_0$. Interpret A^2 in one sentence.

C9.8. Directed graph on nodes $\{1, 2, 3\}$ with adjacency (Boyd: $A_{ij} = 1$ means edge from j to i)

$$A = \begin{bmatrix}0 & 0 & 1\\1 & 0 & 0\\1 & 1 & 0\end{bmatrix}.$$

Given

$$A^2 = \begin{bmatrix}1 & 1 & 0\\0 & 0 & 1\\1 & 0 & 1\end{bmatrix},$$

what does $(A^2)_{31}$ mean? How many length-2 walks from node 1 to node 3?

C9.9. For dynamics $x_{t+1} = Ax_t$, true or false with a one-line reason: $x_5 = A^5 x_0$ and also $x_5 = Ax_4$.

C9.10 (cost). You need $y = (ab^\mathsf{T})c$ for tall vectors of length $n = 1000$. Roughly how many numbers are stored in the intermediate ab^T? Roughly how many in the scalar $b^\mathsf{T}c$? Which plan would you time in software, and why?

Textbook Core

- B 10.3, 10.5, 10.6, 10.7, 10.9, 10.10, 10.13, 10.14, 10.15, 10.22

Stretch

B 10.18, 10.23.

What a quiz item might change

The order of two transformations, matrix dimensions, vectors in the $ab^\mathsf{T}c$ comparison, or timing output from two equivalent computations.